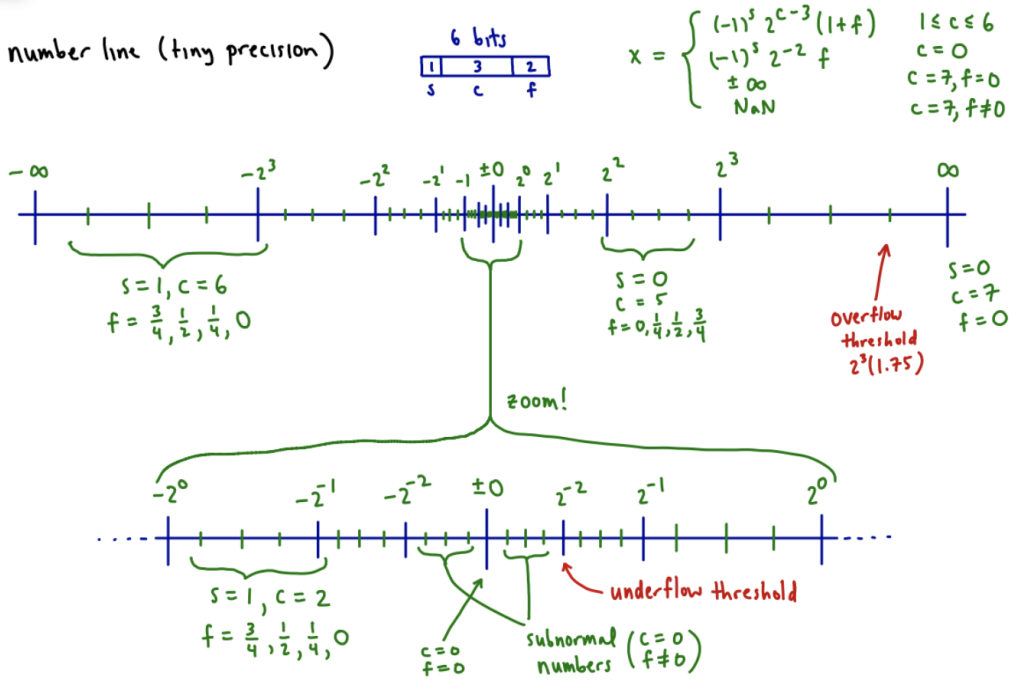
128A Worksheet 2

1 Floating Points

The floating-point number format is one of the greatest works of engineering in the history of computing. Let's walk through why it is the way it is.

Problem 1.1. Explain how the floating-point number format is essentially scientific notation. Explain the role of the bias in the exponent bits.

Problem 1.2. Here is an extremely simplified floating point system: tiny-precision (credit: Prof. Wilkening).



(a) What number is represented by 0 100 10? (Call it x)

(b) What is the exact value of $1/x$?

(c) What is the numerical value of $1/x$ after rounding to the nearest tiny? Express both as a real number and also the binary digits of the tiny format representation.

(d) What is the relative and exact error of the floating point approximation from the previous part?

2 Roundoff Error/Numerical Cancellation

Most commonly, big numerical errors occur when subtracting values of similar magnitude. Sometimes, they occur from adding values of very different magnitudes (but these are usually small errors!). Let's explore how we can create and fix numerical errors.

Problem 2.1. What is "machine epsilon" for a float64?

Problem 2.2. In MATLAB, type in `eps`. Then type in `1 + eps == 1`. Then type in `2 + eps == 2`. Explain what you observed. What is the absolute and relative error of each computation?

Problem 2.3. In MATLAB, run the following:

```
>> x = 1 + 2 ^ (-29)
>> y = 1 + 2 ^ (-30)
>> x^2 - y^2
>> (x - y)(x + y)
```

What is the exact value of `x+y`? Which expression (3 or 4) gets closer to the exact value when evaluated numerically/in MATLAB? Explain why.

Problem 2.4. Consider the function

$$f(x) = \cos(x) - 1$$

near $x = 0$.

(a) Run the following code to plot the default way of writing this function.

```
fplot(@(x)(cos(x)-1),[-1e-7,1e-7])
```

What do you observe?

(b) Can you use a trig identity to create a different way of evaluating $f(x)$ numerically? Try it out, and see if you get a smoother plot.

(c) Once you have a smoother plot, find the worst absolute and relative error made by the first formula compared to your new formula.

(d) Compare the absolute and relative error from part (b). Which numerical error is more problematic?